JMB Lighting — User Manual

JMB one / 4our / 8ight · Firmware 2.x · 2026-08-20 · 4-port shown (1/8-port variants identical unless noted)

1. Front panel

One rotary encoder: **turn** = scroll / adjust, **click** = select, **hold (1.2 s)** = back one level, **keep holding (3 s)** = straight home + lock from any depth. The home screen shows the JMB mark; hold to unlock into the menu. The panel relocks after inactivity (running tests and effects keep going). When no source is decoding, the lock screen reads **No Data** — whatever the CRMX link state — so a linked-but-silent radio can never masquerade as a live source.

Main menu

Item	What it does
Test Mode	Fade / Step RGBW output tests (click cycles, hold exits)
Ports	per-port or All Ports setup, incl. **Presets** — see below
Sources	**the input hub** — wired DMX and CRMX each with live status (the active source + Hz shows inline and in the header); Chain Listen arms here
Health	worst fault inline; Clear Faults + the live universe viewer
Setup	system only: ID, Bluetooth + PIN, curve, encoder, language, protection, cal, reset

Port Settings (per port, or All Ports)

DMX Address and Head Number open the digit editor (turn = digit, click = next, SAVE commits; hold cancels; the footprint hint shows the span). Personality cycles RGB / RGB16 / CCT / CCT16 / **FX10** / **Dim1 / Dim16** (see DMX chart). Effect opens the effect picker. PWM Mode cycles **Smooth 8 kHz / Silent 20 kHz / Camera 62.5 kHz** — Camera is the power-up default and is flicker-free for slow-motion capture (see §8). Current and Temp limits step their trip points and push to the port hardware immediately. Clear Fault re-arms a latched port. **Presets** opens the presets sub-menu (below).

All Ports adds two rows: - **Chain: Local / Global** — Local edits this unit. **Global** also passes a baton down the DMX line after you commit (see §4). - **Apply: Same / Seq / Offset** — how a base address fans out: everyone the same; follow-on by each port's real footprint; or fixed 10-channel spacing (re-personality later without re-patching).

Presets (Port Settings → Presets)

Three rows: - **Dimmer** — a live master brightness. On a single port it opens straight on that port; on **All Ports** it walks each port in turn (turn = level, click = next port). The output follows the encoder **live** — no need to confirm to see it. 1 % steps on a slow turn, coarser when you spin. Hold = cancel and restore the levels you started with. - **Colours** — the factory colour / CCT looks. - **Scenes** — your four user looks (U1–U4).

When Port Settings is on **All Ports** *and* **Chain: Global**, the Dimmer row becomes a **Chain Master**: you set one level and, after a **DEPLOY DIMMER?** confirm, it sets every port on this unit *and* every **armed** unit down the DMX line (see §4). Deploy from the **first** controller in the chain to reach the whole rig.

Sources (the input hub)

One screen answers "what's driving the rig": the header names the **active source and its refresh rate**, and each row carries live status — **DMX In** (Hz or *silent*; click = the line analyser), **CRMX** (*no link / NO DATA /* Hz — data-driven, so "linked but nothing arriving" is visible at a glance; click = link/unlink/antenna) and **Chain Listen** (arms this unit to accept ONE Global baton — always Off at power-up, disarms itself after use).

Setup

Controller ID (shows on screens, in the Bluetooth name after reboot, and in the RDM label) · Bluetooth on/off + PIN · Dimmer Curve (Lin / Soft 1.6 / Thtr 2.2) · Reverse Encoder · Language · Thermal Fold · Sensor Cal · Factory Reset (confirmed).

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2. Output priority

Test mode → Identify (RDM/app) → live DMX → Effect → Manual → hold-last. Live DMX always outranks manual and effects; when the desk vanishes, the last DMX look holds until you set something else. Outputs soft-start over ~1 s at power-up. All sources pass the dimmer curve.

"Live DMX" is the wired input *or* the CRMX radio — they merge into one universe, so everything downstream (patch, FX personality, the diagnostics viewer, RDM) treats the radio exactly like a cable. The header on every screen names the winner: **DMX** or **CRMX**.

2a. Firmware update over Bluetooth (BLE OTA)

The controller updates its own firmware straight over Bluetooth from the app — no USB, no cable, no Wi-Fi.

- **One button** (Setup → *Firmware update* → **Install … over Bluetooth**). The firmware is published alongside the app, so there is no file to find and no way to send a build meant for a different product. The card names the version and build about to be installed, and the build the controller is running, so you can see whether an update would change anything.
- **Checked twice before it runs.** The app verifies the download's size and checksum *before* opening a transfer, and the controller re-checks the checksum across the Bluetooth link before the image is made bootable. A bad download fails in a second rather than after a long upload.
- **Automatic rollback.** The new image boots on probation: if it doesn't come up cleanly and confirm itself, the controller returns to the previous working firmware on its own. The app shows *confirming…* during probation and *firmware confirmed* once the build has proven itself.
- **Manual rollback.** *Roll back to previous* returns to the last firmware deliberately — for a build that runs perfectly well but behaves wrongly, which the automatic check cannot detect. The button stays greyed until there is a previous version stored, which happens after your first update.

- **Port module firmware.** The port processor — the one driving the LEDs and enforcing current, temperature and short-circuit protection — updates the same way, from *Port module firmware* on the same card. Outputs go dark for a few seconds while it is reprogrammed, and its previous version is kept for rollback. If an attempt fails the port simply keeps the firmware it had, and it can be retried over Bluetooth. Do not power the unit off during a port update.

> Keep the app open and the phone unlocked until an update finishes. Outputs hold their last look during a controller update; they go dark during a port module update.
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> Wi-Fi / network DMX (sACN, Art-Net) and Wi-Fi OTA are not part of the shipping firmware — the control path is deliberately kept off the network for responsiveness. Wired DMX, CRMX and BLE cover input and updates.

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3. Effects

Twenty-one built-in dynamic looks per port (Candle, Fire, Fireworks, Water, Siren, Laser, Club, Lightning, Rainbow, Aurora, Sunset, TV Sim, Strobe, Police, Pulse, Party, Traffic, **Fluoro, Paparazzi, Explosion, Clouds**). Set from Port Settings → Effect, the app, or DMX channel 8 of the FX personality. Speed 0–255 (128 = designed tempo) from the app or DMX channel 9. While an effect runs, the app's **RGB sliders become its colour trim** — all at full = the untouched effect; pull one down to take that colour out (Rainbow minus blue, say). The trim is a live tool: it resets to full at power-up. **Effects never start on their own.** When DMX is absent, an effect runs only from the moment you deliberately pick it (panel or app) — a saved effect does not auto-run at power-up, and a desk dropout (cable or CRMX) always **holds the last DMX look** rather than starting one. Pick the effect again after a loss to run it. Siren/Police alternate across ports like a lightbar; Party pulses fully-saturated colour on a 128 BPM bar clock; Fireworks bursts out of true darkness; **Sunset is a one-shot arc** — from golden hour it eases through ember red down to black over ~2 minutes and stays there; re-select it (or dip the port dimmer to 0 and back up) to run another sunset. Traffic plays passing-headlight sweeps with independent timing per port. **The film gags:** Fluoro is a failing fluorescent tube — rapid failed strikes, then it catches and holds with a faint mains buzz; Paparazzi fires irregular hard white pops, independent per port, like a press pit; Explosion is a white-hot flash blooming to an orange fireball with an ember-flicker decay; Clouds drifts daylight slowly dimmer and cooler as cloud crosses the sun, then back.

Rig-wide effect controls (the app's Effects card): **Tap tempo** sets the whole rig's beat-effect speed to the music — tap the app's Tap button in time (or type a BPM, 60–240; 128 is the designed tempo). **Spread** offsets each port in time (0–100 %) so effects sweep across the rig as a wave instead of every port in lockstep. Both persist across reboots.

Capture to scene: the app's Scenes tab saves the current look — **Save current as scene…** (named) or **Capture live** (one tap, auto-named). A scene captures the whole programmer state (colour, or effect + speed), recalls with a crossfade, and rides in the show file. **Up to 100 scenes per phone** (they live in the phone's browser storage, and in any show file you export).

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4. Global addressing (chain baton)

To address a whole chain of controllers from one screen:

1. On **every unit that should take part**: Sources → **Chain Listen** → **ARMED**. (Units left Off are untouched — preset rigs are safe.)
2. On unit 1: Port Settings → All Ports → Chain: **Global**, choose Apply mode, open DMX Address, enter the base, SAVE.
3. Confirm the **DEPLOY CHAIN?** warning. Unit 1 addresses itself, then each armed unit down the line takes the follow-on addresses **and the next controller ID**, shows a receipt, disarms, and relays onward.

Head numbers chain the same way. A unit accepts one baton per arming.

The dimmer chains too. With the same arming in place, Port Settings → All Ports → Chain: Global → **Presets** → **Dimmer** gives a single **Chain Master** level. Set it, confirm **DEPLOY DIMMER?**, and every armed unit down the line takes the same brightness. Only a downstream unit hears the baton, so deploy from the first controller in the chain. (Addresses are untouched — this only moves the dimmer.)

Looks, blackout and full chain too. All Ports → Chain: Global → **Presets** → **Colours** → pick a colour/CCT and confirm **DEPLOY LOOK?** to set that look on every armed unit. Choosing **Off** blacks out the whole chain; **White** takes it to full. The app's **Chain broadcast** card has one-tap **Blackout / Full / DMX In** buttons for the same thing. (Dimmer and looks never re-address or renumber — they only change the output.)

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5. RDM

The controller is a full E1.20 responder: discovery, identify (flashes the ports and strobes the main screen **HERE**), device info, start address, personality — the root device carries the rack view (a SET fans out per the Apply mode) and each port is addressable individually as a sub-device. E1.37-1 dimmer curve and PWM frequency are settable from the desk. The controller also publishes **per-port temperature and current as E1.20 sensors** on the root device ("Port 1 Temp", "Port 1 Current", ...), so any RDM console reads fixture health live. The main screen also flashes the moment discovery finds the unit. The powered DMX thru forwards everything downstream; downstream fixtures' RDM responses are relayed back to the desk automatically.

RDM controller (the app's Diag tab). The controller is also an RDM *controller* for everything plugged into its thru line — a full RDM tester built into the rack. **Scan** discovers every downstream RDM fixture and lists it as a card: model, label, UID, start address, footprint, personality and sensor count. Tap a card to edit it in place — **identify** (flash the fixture to find it), **start address**, **personality**, **label**, and live **sensor reads**. Scanning takes over the output line while it runs, so with an upstream desk live the app asks before taking over; the session releases the thru automatically when you stop (or after 3 minutes idle).

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6. The app

https://app.jmblighting.co.uk/ — runs in **Chrome or Edge** on Android or a desktop (on **iPhone**, open it in the **Bluefy** browser, which supplies the Bluetooth that iOS Safari lacks). Choose **Add to Home screen** for a full-screen, offline-capable app. Tap **Connect**, pick your controller from the list, and enter the PIN (default **0123** — change it in Setup). If a controller has just been reflashed, toggle the phone's Bluetooth off and on once first.

Everything the app sends goes **straight to the controller over Bluetooth** — no account, no cloud, no internet needed once it has loaded, and nothing about your rig leaves the phone. The app also **updates itself**: opened with any connection it loads the newest version, then keeps working from its cached copy at a venue. There is nothing to install from a store and no updates to chase.

Control — the working screen

One card per output. Each card holds the port's **DMX address** and **head number**, its **personality** (RGB or tuneable white, 8- or 16-bit, the 10-channel effect mode, or a **single-channel dimmer** — one DMX level over the port's set colour, ideal for single-colour and IR strips), an **effect** picker with a **speed** control, and the level controls: **R / G / B sliders** — which become **Warm / Cool** on a tuneable-white port (amber and ice-blue tracks, the unused Blue row hidden) — plus two large **jog wheels** for **Dim** and **FX speed** that are easy to ride by feel during a take. Tap the **label** slot in a card's header to name the port (up to 16 characters; the name also appears on that port's own screen when the panel is locked). **Clear all** returns every port to a clean base — full white, effects off, dimmer down — after a confirmation.

Touching any colour or level control **takes manual control automatically** and stops a running effect on that port, so a fader never feels dead. When a live **DMX, sACN or Art-Net** signal is driving the rig, a **"Live DMX in control"** banner appears — your manual moves here are overridden until that signal stops, which is *why* the faders would otherwise seem to do nothing.

Open **Advanced & protection** on a card for that port's safety and quality settings: **PWM mode** (the dimming carrier frequency — see §8), **current limit**, **temperature trip**, **dimmer curve** (**Linear / Soft / Theatre** — how fader travel maps to brightness), and a **Clear fault & re-enable** button that appears only while the port is latched.

Colour — dial a look, don't push channels

A console-style picker for the selected ports (choose **All** or one port at the top). Its five columns — **Dim, Hue, Sat, Kelvin** and **ΔUV** — are all views of one colour: nudge Hue/Sat and Kelvin follows; nudge Kelvin and Hue/Sat follows; the last column you touch wins. **ΔUV** is an independent **green/magenta tint** trim for matching a practical or pleasing a camera. On a tuneable-white port the picker becomes **Kelvin-only** with a live warm/cool mix readout. Reach for this when you want to *choose* a colour rather than balance raw R/G/B by hand.

Scenes — save and recall looks

Store the whole rig's current look as a **scene** and bring it back later, each with its own **crossfade time** (0 = snap). **Capture live** snapshots whatever the controller is outputting right now. Scenes live on the phone and travel inside the exported show file, so a saved show reopens identically on another phone. Tap a scene's name to rename it.

Ports — the health glance

A read-only status page — the same telemetry the port screens show, gathered in one place: each port's **patch**, **output current** (calibrated to within about 1 % of a bench meter), **temperature**, any **fault**, and its active **source**. Your quick "is everything healthy and drawing what I expect?" check.

Diag — the built-in DMX analyser

This is where the controller doubles as a **test instrument** for the signal on the wire. Each tool answers a specific question:

- **Signal card** — *Is a signal arriving, and is it healthy?* Shows whether a signal is present and from where (**DMX / CRMX / sACN / Art-Net**). It reads **CRMX** only when the radio is verifiably carrying the data; otherwise it says simply **DMX**, because wired and wireless arrive on the same input and the source is picked by the XLR socket itself, not by the controller, its **frame rate** with a rolling 90-second graph, **slot count**, **start code**, **time since the last packet**, alternate-start-code packet count, and **error counts**. Your first check that the desk is actually reaching the rig at a good rate.

- **Universe analyser** — *What is every channel doing?* Opens full-screen and lays out the whole universe as **numbered channel meters** — the channel number, a live level bar and its value — so you can read which channel is doing what at a glance and scroll the full 512. Switch between **Live**, **Min**, **Max** and **Activity** views; **Start capture** records the highest and lowest level each channel reaches over a take (that feeds the Min/Max views), and **Activity** highlights which channels are moving. Tap any channel for its detail: level, percentage, the **port it patches to**, its captured range, and how many times it has changed.

- **Flicker finder** — *Which channel is causing a flicker?* The intermittent-flicker hunter. **Arm** it and every single channel-value change on the wire is logged with the old value, the new value and how long ago it happened — so a light that flickers once every few minutes, or a desk quietly stepping a level, is **caught and timestamped** instead of leaving you guessing. It keeps logging even if you leave the tab, so an intermittent fault can be pinned to the exact channel and moment it occurred.

- **RDM universe** — *What's downstream, and can I fix it from here?* Discovers and controls every RDM fixture down the line like a dedicated RDM tester: **Scan** to find them, then read and set each fixture's **address**, **personality**, **label**, **Identify** and **sensor** readings (see §5). Scanning takes over the output line while it runs.

- **Port bus** — a health table for the internal link to each output module (RX count, CRC errors, fault state), for diagnosing a port that has stopped responding.

Setup — controller and housekeeping

Controller **ID** and **Bluetooth name** (what the pairing list shows), **apply mode** (how a change made to "All" fans out across the ports), **CRMX wireless** (link / unlink / antenna select — the status reads **green when linked**), **Firmware update over Bluetooth** (§2a), **Flash to find** (identifies this controller for 5 s so you can pick it out of a rack: units with a front panel strobe their screen and leave the rig untouched; the screenless **JMB one** blinks its own outputs instead, being the only thing it can show you with), **Reset all ports** with an

Undo, **show-file export / import** (one **.jmb** file carrying the whole rig — addresses, heads, labels, looks and scenes — reloadable on any phone or controller), **change PIN**, and **Disconnect**.

Test sequences (Setup, top of the tab) drive every port directly, overriding every source — **live DMX included** — until switched **Off**. A banner appears on the Control tab while a test runs, with its own *Stop test* button, and the **Off** button lights when nothing is running so you can tell at a glance.

- **Step RGBW** — full drive red → green → blue → white, one second each, gamma bypassed. Every channel at maximum and identical on all ports, so a dead channel or a weak output stage stands out immediately.
- **Fade** — a smooth rainbow sweep, for judging dimming smoothness and colour blending.

The wiring check runs automatically during Step RGBW. Because that test lights one colour at a time, and each port measures its own current, the reading during each step *is* that channel's current. After one full four-second cycle the app lists them per port:

`Port 1: R 1.90A · G 1.60A · B 1.50A · W 4.90A ✓ consistent`

Colours legitimately differ — red's forward voltage is about a volt below green and blue — so the check only speaks up for unambiguous cases: an output drawing nothing, an output drawing markedly more than its siblings, or the two together, which means two colours are sharing a terminal (see §7). The figures stay on screen after you stop the test.

Effects on tuneable-white ports. On a CW/WW personality the effect list only offers the effects that read on a white strip (Candle, Fire, Lightning, TV Sim, Strobe, Pulse, Fluoro, Paparazzi, Explosion); the colour-only effects are hidden so you can't pick one that would do nothing.

Demo mode. *Explore in demo mode* on the connect screen walks the entire app — every tab, the analyser, even an RDM scan that discovers example fixtures — with no controller present. Nothing a demo user touches goes anywhere; it is for learning the app or showing it off before the hardware is in front of you.

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7. Fault handling

Ports protect themselves: over-current (hiccup ×3 then latch), bolted-short µs trip, over-temperature, ground-leak imbalance, comms-loss failsafe. Latched ports show a flashing fault banner on their screen, the fault word appears in the menu and the app, and Clear Faults / Port Reset re-arms.

Wiring fault. Separate from the electrical faults above, and raised by the controller rather than the port. Running **Step RGBW** measures each output's current one colour at a time; if one output draws roughly double while another draws nothing, two colours are almost certainly sharing a terminal. The port is **inhibited** and both the panel and the app name the pair — *P1 WIRE R+B* means the blue load is on the red terminal.

This matters because it is the one wiring mistake the port's own protection cannot see: the current still leaves and returns through the same loop, so over-current and ground-leak detection are both satisfied while one output quietly carries two channels' worth. **Clear Faults** on the panel, or *Clear wiring fault & re-enable* in the app, restores the port; if the wiring is still crossed it latches again the next time you run the test. An output with simply *nothing* connected is reported but never inhibits a port — a tuneable-white fixture legitimately has an unused output.

Thermal foldback (Setup → Thermal Fold, on by default): before a port hits its over-temp trip, it eases the output down over the last 10 °C (to a 30 % floor) so a hot fixture *rides through* instead of hard-latching mid-show. The latch stays as the real-fault backstop.

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8. Camera & slow-motion work

Each port dims with a high-frequency PWM carrier, set per port under **Port Settings** → **PWM Mode** (also in the app, and over RDM via MODULATION_FREQUENCY):

Mode	Carrier	Use it for
Camera *(default)*	**62.5 kHz**	Any shoot — flicker-free on high-speed cameras
Silent	20 kHz	Quietest drive (no coil whine) when no camera is present
Smooth	8 kHz	Maximum low-end dimming smoothness, eye-only / theatre

All three carry the full ~16-bit dimming resolution (temporal dithering makes up the difference at the higher carriers), so the picture is identically smooth — only the flicker frequency changes.

Setting up for slow-motion / high-speed capture:

- **Leave every port on Camera (62.5 kHz)** — it's the power-up default. At 62.5 kHz the output is flat through exposures short enough for normal high-speed work (comfortably to ~1000 fps at sensible shutter angles).
- **Keep all ports on the same mode** so multiple fixtures match on camera. Use All Ports → PWM Mode, or set it once over RDM, to change the whole rig together.
- **Extreme frame rates** (multi-thousand fps) with very short exposures: shoot a quick test first. *Any* PWM dimmer can show faint banding when the exposure catches only a few carrier cycles — if you see it, widen the shutter angle or drop the frame rate slightly. Higher isn't selectable beyond 62.5 kHz.
- **Turn strobe looks off for clean capture.** The Strobe / Police effects and the FX personality's shutter channel (CH10) *deliberately* chop the output — that's the effect, not carrier flicker. Set the shutter open / effect Off.
- **Fades and colour moves are flicker-free** at the carrier; dial fade times and the dimmer curve as normal — nothing special needed for the camera.

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9. Quick reference

- Default PIN: **0123** · default addresses: sequential RGB
- DMX chart: see `docs/DMX_CHART.md`
- Start address caps at 513 − footprint so the whole personality fits the universe (FX 10ch → max 503); it drops automatically if you switch a port to a larger personality
- Chain Listen is always Off after power-up — arming is deliberate
- The app's Undo restores the state captured before the last Reset-all

- Supply: 24 V standard (12 V supported — match the strip voltage). All protections are current- and temperature-based, so they work identically on either rail; at 12 V each port delivers half the wattage for the same current limit